

Solute, Solvent & Solution

Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Supporting vocabulary

Name: _____ Date: _____

LEARNING INTENTION & SUCCESS CRITERIA

I can identify the parts of a solution and explain dissolving.

I can name the solute, solvent and solution; distinguish dissolving from melting; and explain how a dissolved solid can be recovered.

KEY CONCEPTS

Solute: a substance that dissolves. Solvent: the substance that dissolves it. Solution: the evenly mixed mixture formed.

Soluble: able to dissolve in a particular solvent. Insoluble: unable to dissolve in that solvent under the conditions tested.

WORKED EXAMPLE

A small amount of salt dissolves in water: salt = solute; water = solvent; saltwater = solution. The salt is still present even when its crystals are no longer visible.

PART A - WARM UP

1. Sugar dissolves in water. Name the solute, solvent and solution.

2. Sand does not dissolve in water. Is sand soluble or insoluble in water?

3. Ice becomes liquid water. Is this melting or dissolving? Explain.

Solute, Solvent & Solution

Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Apply & explain

Name: _____ Date: _____

PART B - CORE SKILLS

4. Compare sugar in water and sand in water. Which can form a solution? Explain the difference you would observe.

5. Mia says, "The salt disappeared, so the saltwater contains only water." Explain why she is incorrect.

6. A cup of water has a mass of 100 g. You add 5 g of salt. All the salt dissolves and nothing spills or evaporates. What is the total mass? Explain.

PART C - CHALLENGE

7. How could you recover salt from saltwater? Explain what happens to the water and what remains. Do not heat anything without adult supervision.

REFLECTION & I CAN CHECK

One idea I corrected today: _____

I can identify solution parts: Yes / Not yet

I can explain why dissolved salt is still present: Yes / Not yet

Solute, Solvent & Solution

Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Parent / teacher answers

Name: _____ Date: _____

ANSWER KEY

1. Solute: sugar. Solvent: water. Solution: sugar solution/sugar water.
2. Insoluble in water under these conditions.
3. Melting: solid water changes to liquid water. It is not a substance dissolving in a solvent.
4. Sugar can form a solution if the amount is within what the water can dissolve. Sand stays as visible grains and may settle. Sugar crystals become no longer visible.
5. Salt is distributed through the water; being invisible does not mean it is absent. Recovering salt by evaporation gives evidence.
6. 105 g. Dissolving does not remove the salt's mass.
7. Allow water to evaporate from a shallow dish. Water enters the air as water vapour; salt remains. This recovers salt, not the water.

TEACHING & PARENT TIP

Use small amounts of salt, sugar and sand in separate water samples. Compare observations without tasting. A solution can be coloured and still be transparent: “clear” does not always mean colourless or pure.

Ask: “What evidence would show the salt is still there?” Connect recovery to reversible change. Solute/solvent vocabulary supports Year 6 learning; detailed mixture classification develops further in Year 7.

TEACHER FEEDBACK

Strength: _____

Next step: _____